

Many LSAs are equipped with water-cooled engines. LSAs may be tightly cowled, which reduces drag. A liquid-cooled engine minimizes the need for cylinder cooling inlets, which further reduces drag and improves performance. This does present a new system for a transitioning pilot to check. Preflighting this system requires that the radiator, coolant hoses, and expansion tank are checked for condition, freedom from leaks, and coolant level requirements.

Split flaps may be used on some LSA designs. [Figure 17-14] These flaps hinge down from underneath the wing and inspecting these flaps requires the pilot to crouch and twist low for inspection. A suitable handheld mirror can facilitate inspection without undue twisting and bending. In an attempt to keep complexity to a minimum, flap control is typically a handle that actuates the flaps. A pilot should verify that the flaps extend and retract smoothly.



Figure 17-14. *Split flap.*

Before Start and Starting Engine

Once a pilot has completed the preflight inspection of the LSA, the pilot should properly seat themselves in the airplane ensuring that the rudder pedals can be exercised with full-range movement without over-reaching. Seat belts should be checked for proper position and security. The pilot should continue to use the POH checklists. Starting newer generation LSA engines can be quite simple and only require the pull of the choke and a twist of the ignition switch. If the LSA is equipped with a standard certificated engine, starting procedures are normal and routine. The canopy or doors of an LSA may have quite different latching mechanisms than standard airworthiness airplanes. Practice latching and unlatching the doors or canopy to ensure that understanding is complete. Having a gull-wing door or sliding canopy “pop” open in flight can become an emergency in seconds.

Taxi

LSAs may have a full-castoring or steerable nose-wheel, or a tailwheel if equipped with conventional gear. In order to taxi a full-castoring nose-wheel equipped airplane, the use of differential brakes is required. This type of nose-wheel can require practice to keep the airplane on the centerline while minimizing brake application or damage to the tires. If the taxi speed is too slow, application of a brake can cause the aircraft to pivot to a stop, rather than adjust in direction. This results in excessive brake and tire wear. If the speed is too fast, excessive brake wear is likely.

Transitioning into an LSA with conventional gear (tailwheel) should occur initially during no-wind conditions. Due to its light weight the airplane, requires the development of the proper flight control responses before operations in any substantial wind.

Takeoff and Climb

Takeoff and climb performance of LSAs can be spirited as they typically have a high horsepower-to-weight ratio and accelerate quickly. Due to design requirements for low stall speeds, LSAs typically have low rotation and climb speeds with impressive climb rates. Like other airplanes, the pilot should be flying the published speeds as given in the airplane’s POH. Stick (yoke or stoke) forces tend to be light, which may lead a transitioning pilot to initially over-control. The key is to relax, have reasonable patience, and input only appropriate flight control pressures needed to get the required response. If a transitioning pilot is inducing excessive control inputs, they should minimize flight control pressures, set attitudes based on outside references, and allow the airplane to settle.

During climbs, visibility over the nose may be difficult in some LSAs. As always, it is important to properly clear the airspace for traffic and other hazards. Occasionally lowering the airplane’s nose to get a good look out toward the horizon is important for managing flight safety. Shallow banked turns in both directions of 10° to 20° also allow for clearing. Because flight control pressures tend to be light, it is easy to get in the habit of flying with a light-sport airplane out of trim. This is to be avoided. Trim off any flight control pressures. This allows the pilot to focus as much time as possible looking outside.

Cruise

After leveling off at cruise altitude, the airplane should be allowed to accelerate to cruise speed, reduce power to cruise rpm, adjust pitch, and then trim off any flight control pressures. [Figure 17-15] The first time a transitioning pilot sees cruise rpm setting of 4,800 rpm (or as recommended), they may have a sense that the engine is turning too fast; however, remember that the engine has gear-reduction drive and the propeller is turning much slower. If the LSA is equipped with a standard aircraft engine, rpm should be in a range comparable to airplanes the transitioning pilot is used to. The pilot should refer to the Cruise Checklist to ensure that the airplane is properly configured.

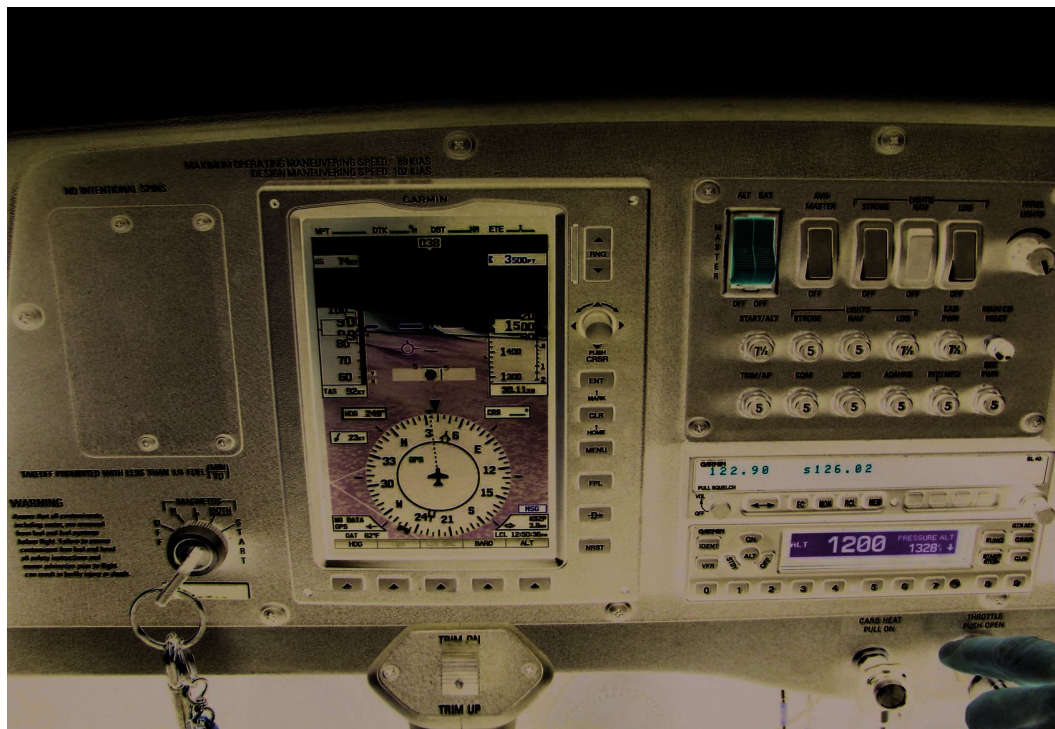


Figure 17-15. EFIS indication of level cruise flight.

In slower cruise flight, stick forces are likely to be light; therefore, correction to pitch and roll attitudes should be made with light pressures. Excess pressure used to correct a deviation may cause a series of pilot-induced oscillations. The pilot should use fingertip pressures only and not use a wrapped palm of the hand. Stick forces can change dramatically as airspeed changes. For example, what could be considered light control pressures at 80 knots may become quite stiff at 100 knots. A flight instructor-S or flight instructor-A experienced in the light-sport airplane is able to demonstrate this effect, which is dependent on the specific model of LSA.

LSA maneuvers such as steep turns, slow flight, and stalls are typical. These maneuvers should be practiced as part of a good transition training program. Steep turns in LSA airplanes tend to be quite easy to perform with precision. Light flight control pressures, stick mounted trim (if installed), and highly differential ailerons (if part of the airplane's design), make the maneuvers seem simpler than in heavier airplanes. Basic aerodynamics applies to any airplane. Factors, such as over-banking tendency, are still prevalent and should be expected.

Slow flight in LSAs is accomplished at slower airspeeds than standard airworthiness airplanes since stall speeds tend to be well below the 45-knot limit. Practicing slow flight demonstrates the unique capability of LSAs. Recovery from power-off stalls involves lowering the nose. Application of power puts the airplane back flying. However, a pilot should understand that control pressures tend to be light, and an aggressive forward movement of the elevator is generally not required. In addition, proper application of rudder to compensate for propeller forces is required, and retraction of any flap should be completed prior to reaching VFE, which occurs quickly if full power and nose down pitch attitude are maintained. Power-on stalls can result in a very high nose-up attitude unless the airplane is adequately slowed down prior to the maneuver. In addition, some manufacturers limit pitch attitudes to 30° during power-on stalls. If aggressive pitch attitudes are coupled with uncoordinated rudder inputs, spin entry is likely to be quick and aggressive.

Depending on the LSA design, especially those airplanes which use control tubes rather than wires and pulleys, flight in turbulence may couple motion to the stick rather distinctively. If a transitioning pilot's flight experience is only with airplanes that have control cables and pulleys, the first flight in turbulence may be disconcerting; however, once the pilot becomes familiar with the control sensations induced by the turbulence, it only becomes another means for the pilot to feel the airplane.

Approach and Landing

Approach and landing in an LSA is routine and comfortable. Speeds in the pattern tend to be in the 60-knot range. Flap limit airspeeds tend to be lower in LSAs than standard airworthiness airplanes, so managing airspeed is important. Light control forces require smooth application of control pressures to avoid over-controlling. Pitch and power are the same in an LSA as in a standard airworthiness airplane.

The weight limit of light-sport airplanes makes crosswind landings an important subject to focus on. The pilot should realize that strong gusty crosswind conditions may exceed the airplane's control capability resulting in loss of control during the landing. Manufacturers place a maximum demonstrated crosswind speed in the POH, and until sufficient practice and experience is gained in the airplane, a transitioning pilot should have personal minimums that do not approach the manufacturer's demonstrated crosswind speed. Control application does not change for crosswind technique in an LSA. However, the LSA's weight, slow landing speeds, and light control forces can result in a pilot making control deflections that exceed those necessary to compensate for the crosswind.

Emergencies

While an LSA is designed to be simple, a complete knowledge of its systems is needed such that a transitioning pilot is able to respond properly to any emergency.

The airplane's POH describes the appropriate responses to the various emergency situations that may be encountered. [Figure 17-16] Consider a few examples: the EFIS is displaying a red "X" across the airspeed tape, electric trim runaway, or control system failure. The pilot should be able to respond to immediate action items from memory and locate emergency procedures quickly. In the example of trim runaway, the pilot needs to quickly assess the trim runaway condition, locate and depress the trim disconnect (if installed), or pull the trim power circuit breaker. Then depending on control forces required to maintain pitch attitude, the pilot may need to make a no-flap landing due to the flap pitching moments. If the EFIS "blanks" out and POH recovery procedures do not reset the EFIS, an LSA pilot may have to be prepared to land without airspeed, altitude, or vertical speed information. An effective training program covers these emergency procedures.

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■ Section 3 provides checklist and amplified procedures for coping with emergencies that may occur. Emergencies caused by airplane or engine malfunctions are extremely rare if proper preflight inspections and maintenance are practiced. Enroute weather emergencies can be minimized or eliminated by careful flight planning and good judgment when unexpected weather is encountered. However, should an emergency arise, the basic guidelines described in this section should be considered and applied as necessary to correct the problem. In any emergency situation, the most important task is continued control of the airplane and maneuver to execute a successful landing.	
Emergency procedures associated with optional or supplemental equipment are found in Section 9, Supplements.	
AIRSPEEDS FOR EMERGENCY OPERATIONS	
ENGINE FAILURE AFTER TAKEOFF	
Wing flaps UP.....	70 KIAS
Wing flaps 10" - FULL	65 KIAS
■ MAXIMUM OPERATING MANEUVERING SPEED	
1320 pounds.....	89 KIAS
1200 pounds.....	85 KIAS
1100 pounds.....	80 KIAS
■ DESIGN MANEUVERING SPEED	
	102 KIAS
MAXIMUM GLIDE.....	
	70 KIAS
PRECAUTIONARY LANDING WITH ENGINE POWER.....	
	60 kias
LANDING WITHOUT ENGINE POWER	
Wing flaps UP.....	70 KIAS
Wing flaps 10" - FULL	65 KIAS

Figure 17-16. Example of a POH Emergency Procedures section.

Post-Flight

After the airplane has been shutdown, tied-down, and secured, the pilot should conduct a complete post-flight inspection. Any squawks or discrepancies should be noted and reported to maintenance. Transitioning pilots should insist on a training debriefing where critique and planning for the next lesson takes place. Documentation of the pilot's progress should be noted in the student's records.

Key Points

LSAs with an open flight deck, easy build characteristics, low cost, and simplicity of operation and maintenance tend to be less aerodynamic and incur more drag. When combined with their low mass and inertia, these LSAs tend to decelerate rapidly when power is reduced. When attempting a crosswind landing in a high-drag LSA, a rapid reduction in airspeed prior to touchdown may result in a loss of rudder and/or aileron control, which may push the aircraft off of the runway heading. To avoid loss of control, maintain airspeed during the approach. When power is reduced, it may be necessary to lower the nose of the aircraft to a fairly low pitch attitude in order to maintain airspeed.

If the pilot makes a power-off approach to landing, the approach angle will be high and the landing flare will need to be close to the ground with minimum float. This is because the aircraft will lose airspeed quickly in the flare and will not float like a more efficiently designed aircraft. Too low of an airspeed during the landing flare may lead to insufficient energy to arrest the descent and may result in a hard landing. Maintaining power during the approach will result in a reduced angle of attack and will extend the landing flare allowing more time to make adjustments to the aircraft during the landing. Always remember that rapid power reductions require an equally rapid reduction in pitch attitude to maintain airspeed.

In the event of an engine failure in an LSA, quickly transition to the required nose-down flight attitude in order to maintain airspeed. For example, if the aircraft has a power-off glide angle of 30 degrees below the horizon, position the aircraft to a nose-down 30 degree attitude as quickly as possible. The higher the pitch attitude is when the engine failure occurs, the quicker the aircraft will lose airspeed and the more likely the aircraft is to stall. Should a stall occur, decrease the aircraft's pitch attitude rapidly in order to increase airspeed to allow for a recovery. Stalls that occur at low altitudes are especially dangerous because the closer to the ground the stall occurs, the less time there is to recover. For this reason, when climbing at a low altitude, excessive pitch attitude is discouraged.

Chapter Summary

LSAs are a category of small, lightweight aircraft that may include advanced systems, such as parachutes, EFIS, and composite construction. While the transition is not difficult, it does require a properly designed transition training program led by a competent flight instructor-S or flight instructor-A. Safety is of utmost importance when it comes to any flight activity. In order to properly assess the hazards of flight and mitigate flight risk, a pilot needs to develop the appropriate knowledge, risk management, and skill, to effectively and safely pilot an LSA.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 18: Emergency Procedures

Introduction

This chapter describes certain abnormal and emergency situations that may occur in flight. The key to successful management of an emergency situation, and/or preventing an abnormal situation from progressing into a true emergency, is a thorough familiarity with, and adherence to, the procedures developed by the airplane manufacturer. The following guidelines are generic and are not meant to replace the airplane manufacturer's recommended procedures contained in the Federal Aviation Administration (FAA) approved Airplane Flight Manual and/or Pilot's Operating Handbook (AFM/POH). Rather, they are meant to enhance the pilot's general knowledge in the area of abnormal and emergency operations. If any of the guidance in this chapter conflicts in any way with the manufacturer's recommended procedures for a particular make and model airplane, the manufacturer's recommended procedures take precedence.

Emergency Landings

This section contains information on emergency landing techniques in small fixed-wing airplanes. The guidelines that are presented apply to the more adverse terrain conditions for which no practical training is possible. The objective is to instill in the pilot the knowledge that almost any terrain can be considered "suitable" for a survivable crash landing if the pilot knows how to use the airplane structure for self-protection and the protection of passengers.

Types of Emergency Landings

The different types of emergency landings are defined as follows:

- Forced landing—an immediate landing, on or off an airport, necessitated by the inability to continue further flight. A typical example of which is an airplane forced down by engine failure.
- Precautionary landing—a premeditated landing, on or off an airport, when further flight is possible but inadvisable. Examples of conditions that may call for a precautionary landing include deteriorating weather, being lost, fuel shortage, and gradually developing engine trouble.
- Ditching—a forced or precautionary landing on water.

A precautionary landing, generally, is less hazardous than a forced landing because the pilot has more time for terrain selection and the planning of the approach. In addition, the pilot can use power to compensate for errors in judgment or technique. The pilot should be aware that too many situations calling for a precautionary landing are allowed to develop into immediate forced landings, when the pilot uses wishful thinking instead of reason, especially when dealing with a self-inflicted predicament. The non-instrument-rated pilot trapped by weather, or the pilot facing imminent fuel exhaustion who does not give any thought to the feasibility of a precautionary landing, accepts an extremely hazardous alternative.

Psychological Hazards

There are several factors that may interfere with a pilot's ability to act promptly and properly when faced with an emergency. Some of these factors are listed below.

- Reluctance to accept the emergency situation—a pilot who allows the mind to become paralyzed at the thought that the airplane will be on the ground in a very short time, regardless of the pilot's actions or hopes, is severely handicapped in the handling of the emergency. An unconscious desire to delay the dreaded moment may lead to such errors as: failure to lower the nose to maintain flying speed, delay in the selection of the most suitable landing area within reach, and indecision in general. Desperate attempts to correct whatever went wrong at the expense of airplane control fall into the same category.
- Undue concern about getting hurt—fear is a vital part of the self-preservation mechanism. However, when fear leads to panic, we invite that which we want most to avoid. The survival records favor pilots who maintain their composure and know how to apply the general concepts and procedures that have been developed through the years. The success of an emergency landing is as much a matter of the mind as of skills.